Executive Summary

Week 3 • Healthcare Operations & Financial Intelligence Dashboard • Track A: Data Analytics

This summary covers what the encounter, procedure and payer data show about how the hospital is running: how busy it is, how long patients stay, and who ends up paying. The figures come from the Power BI dashboard, which is built on 27,891 recorded encounters.

27,891	$101.51M	$31.10M	$70.42M	58
Total encounters	Total claim cost	Payer coverage	Patient responsibility	Long-stay visits

Key findings

1. **Volume peaked in 2014 and 2021.** There were 3,885 encounters in 2014 and 3,530 in 2021. 2022 shows only 220, which is too low to be a real drop. It most likely reflects incomplete data, so I do not treat it as a decline.
2. **Ambulatory care carries the workload.** Ambulatory visits make up 44.95% of encounters, or 12,537 visits.
3. **Long stays are rare.** Only 58 of 27,891 encounters (0.21%) lasted longer than 24 hours. In this data, extended stays are not what is putting pressure on the hospital.
4. **Payers covered under a third of what was billed.** Total claim costs were $101.51M and payers covered $31.10M. That leaves $70.42M, about 69%, as patient responsibility.
5. **Nearly half of encounters had no payer coverage at all.** 13,586 encounters (about 49%) have zero recorded coverage. The NO_INSURANCE category is the biggest source, with 8,807 visits and over $49.26M in claims. That one category accounts for roughly 70% of the patient responsibility total.

How the $101.51M in total claim costs splits between payers and patients

Recommendations

1. Screen uninsured patients for coverage at registration

Finding	NO_INSURANCE accounts for 8,807 visits and over $49.26M in claims with no coverage, about 70% of the patient responsibility total.
Action	Add coverage and eligibility screening at registration, and offer financial counselling and payment plans to patients without cover.
Benefit	Moves some claims from no coverage to covered and reduces the amount the hospital has to recover from patients.
Limitation	The data shows billed claims, not money collected. It also cannot confirm whether NO_INSURANCE means truly uninsured or simply not recorded.

2. Review zero-coverage claims that have a named payer

Finding	13,586 encounters have zero payer coverage, but only 8,807 of them are NO_INSURANCE. That leaves about 4,800 encounters where a payer is recorded and still paid nothing.
Action	Sample these encounters to find out whether claims were denied, never submitted or poorly recorded, then take the patterns into payer contract discussions.
Benefit	Recovers revenue that may be owed and gives finance a stronger position with payers.
Limitation	The dataset gives no denial reasons, so finance would need claims records to explain these cases. The 4,800 figure also assumes every NO_INSURANCE encounter has zero coverage.

3. Plan staffing and capacity around ambulatory demand

Finding	Ambulatory visits are 44.95% of encounters, while only 0.21% run past 24 hours.
Action	Put scheduling, staffing and room capacity for ambulatory services first, ahead of planning for extended stays.
Benefit	Directs resources to where most of the demand is.
Limitation	These files hold no staffing or capacity data, and the 2022 figures look incomplete. The 2022 data should be confirmed with the source before the trend is used for forecasting.

Read these figures with care

- Patient responsibility is total claim cost minus payer coverage. It may include amounts that were written off or paid in another way.
- About 70% of encounters have no reason recorded, so I did not break results down by reason.